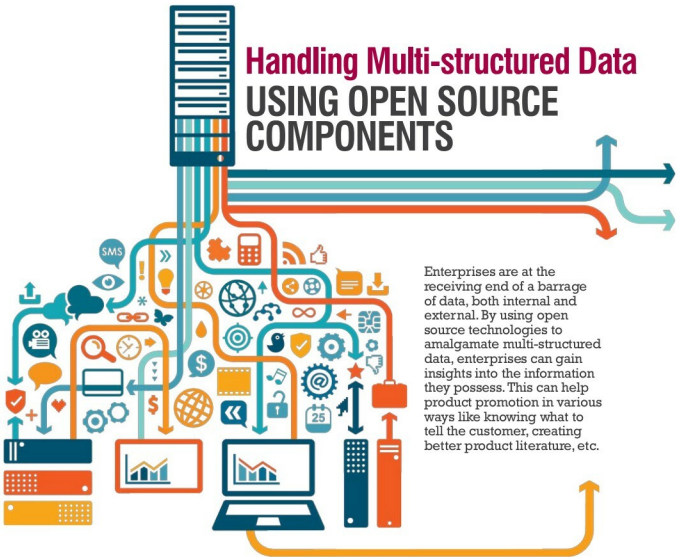




Handling Multi-structured Data USING OPEN SOURCE COMPONENTS

Enterprises are at the receiving end of a barrage of data, both internal and external. By using open source technologies to amalgamate multi-structured data, enterprises can gain insights into the information they possess. This can help product promotion in various ways like knowing what to tell the customer, creating better product literature, etc.

Today, an enterprise has data from a variety of sources, with internal data coming from local infrastructure, home grown applications, databases, enterprise resource planning (ERP) systems and customer relationship management (CRM) systems. Enterprises collect data from external sources such as social media and blogs to analyse end users' comments on their products and services, and how these fare with respect to competition. The data is multi-structured. Firms face the huge challenge associated with the volume, velocity and variety of data.

Making sense of this data deluge requires careful harnessing and analytical tools to gain the right insights. An opensource platform can typically provide the much needed diversity to combine data exploration, data model extraction and discovery of relationships. This can be used to come with up a context based approach for sentiment analysis.

In this article, we propose an open source stack that can serve as a building block to handle multi-structured data that assists data scientists to get actionable insights and suggest

recommendations. The actionable insights could be on purchasing behaviour, product sentiments, services offered by the enterprise and their quality. We will also provide details of a use case from the retail industry and how this can be extended to other verticals.

The platform for multi-structured data handling

Let us briefly cover the components of the open source stack that we will be using for handling multi-structured data.

ELK: This is an open source software stack that combines Elasticsearch, Logstash and Kibana, and is primarily geared to handle multi-structured data to deliver real-time insights. Elasticsearch is an indexer. It has the built-in capability to perform basic and advanced search besides analysis of the data. Logstash collects various types of data, and has the ability to parse and format the data. Further, it sends this information to applications such as Elasticsearch for indexing the transformed data. Kibana is a Web based application that helps in visualisation and analysis of the data. It helps in